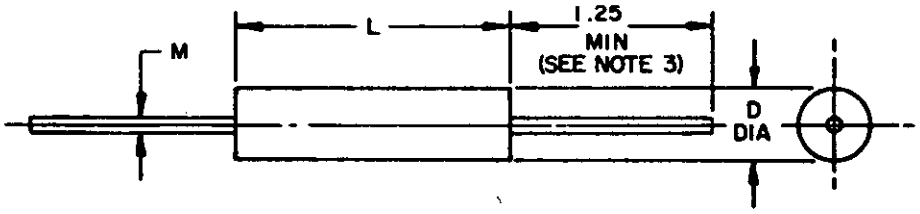
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), ESTABLISHED RELIABILITY, STYLES CKR20, CKR17, AND CKR18

This specification is approved for use by all Departments and Agencies of the Department of Defense.

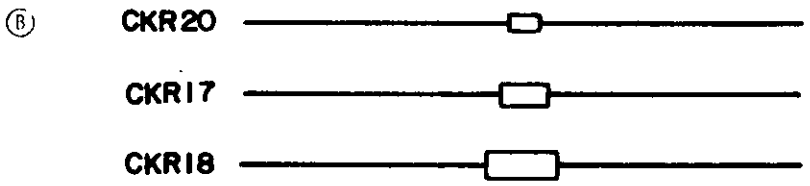
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39014.

FAILURE RATE LEVEL "L" IS INACTIVE FOR NEW DESIGN AFTER



Style	Dimensions		
	L	D	M (dia)
CKR20	.160 ±.010	.090 ±.010	.019 ±.002
CKR17	.250 ±.015	.090 ±.010	.019 ±.002
CKR18	.390 ±.015	.145 ±.015	.025 ±.002

INCHES	MM	INCHES	MM
.002	.05	.090	2.29
.007	.18	.145	3.68
.010	.25	.160	4.06
.015	.38	.250	6.35
.019	.48	.390	9.91
.025	.64	1.25	31.8
.030	.76		



APPROX ACTUAL SIZE

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Lead length may be a minimum of one inch long for use in tape and reel automatic insertion equipment, when specified.

Ⓑ denotes changes.

FSC 5910

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REQUIREMENTS:

Dimensions and configuration: See figure.

Case type: Tubular, glass or ceramic, hermetic seal.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: See table I.

Dissipation factor: Not more than 2.5 percent.

Insulation resistance:
At 25°C - Not less than 100,000 megohms or 1,000 megohm-microfarads, whichever is less.
At 125°C - Not less than 10,000 megohms or 100 megohm-microfarads, whichever is less.

Seal (styles CKR20, CKR17, and CKR 18): In accordance with MIL-C-39014.

Voltage-temperature limits: In accordance with MIL-C-39014.

Capacitance change with reference to 25°C	
Steps A through D of table V Bias = 0 volts	Steps E through G of table V Bias = rated voltage
±15 percent	+15, -25 percent

Immersion:
IR - Not less than initial 25°C requirement.

Salt spray (corrosion): Not applicable

Moisture resistance:
IR - Not less than initial 25°C requirement.

Resistance to soldering heat:
ΔC - +20, -0 percent from initial measured value.

Life:
Rated conditions, 100 percent of dc rated voltage at 125°C.
0 through 3,000 hours:
IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC - ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.
4,000 through 32,000 hours:
IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC - ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.
Accelerated condition, 200 percent of dc rated voltage at 125°C.
0 and 250 hours:
IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC - ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.

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1,000, 2,000 and 4,000 hours:
IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC - ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.

Marking: In accordance with MIL-C-39014 except that minimum marking as shown in the following examples will be permitted for styles CKR20 and CKR17. Full marking shall be included on the package.

CKR20

JTM - J (for JAN), and trademark.
100 - Capacitance 1/.
KP8 - Capacitance tolerance 2/, FR level, and date code (year).
31A - Lot code (1 week intervals), and inspection lot code.

CKR17

JCK17 - J (for JAN), and abbreviated style.
103KP - Capacitance 1/, capacitance tolerance 2/, and FR level.
(B) 7844A - Date code, and lot code.
TM - Manufacturer's two or three digit trademark or symbol.

1/ The capacitance will be shown by three digits; the first two digits shall be significant and the third digit shall be the multiplier.
2/ The capacitance tolerance will be shown by the letter K (±10%) or M (±20%).

Part number: M39014/13- (Dash number from table I).

Supersession data: See table II.

(B) TABLE I. Dash number and electrical characteristics.

Dash number				Capacitance	Capacitance tolerance	DC rated voltage
Failure rate level (%/1,000 hours)						
1.0(M)	0.1(P)	0.01(R)	0.001(S)			
Style CKR20, voltage-temperature limits of ±15% and +15, -25%						
				pF	±percent	volts
1101	1201	1301	1401	10	10	100
1102	1202	1302	1402	10	20	100
1103	1203	1303	1403	12	10	100
1104	1204	1304	1404	15	10	100
1105	1205	1305	1405	15	20	100
1106	1206	1306	1406	18	10	100
1107	1207	1307	1407	22	10	100
1108	1208	1308	1408	22	20	100
1109	1209	1309	1409	27	10	100
1110	1210	1310	1410	33	10	100
1111	1211	1311	1411	33	20	100
1112	1212	1312	1412	39	10	100
1113	1213	1313	1413	47	10	100
1114	1214	1314	1414	47	20	100
1115	1215	1315	1415	56	10	100
1116	1216	1316	1416	68	10	100
1117	1217	1317	1417	68	20	100
1118	1218	1318	1418	82	10	100
1119	1219	1319	1419	100	10	100
1120	1220	1320	1420	100	20	100
1121	1221	1321	1421	120	10	100
1122	1222	1322	1422	150	10	100
1123	1223	1323	1423	150	20	100
1124	1224	1324	1424	180	10	100
1125	1225	1325	1425	220	10	100
1126	1226	1326	1426	220	20	100

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(B) TABLE I. Dash number and electrical characteristics - Continued.

Dash number				Capacitance	Capacitance tolerance	DC rated voltage
Failure rate level (%/1,000 hours)						
1.0(M)	0.1(P)	0.01(R)	0.001(S)			
Style CKR20, voltage-temperature limits of ±15% and +15, -25%						
				pF	±percent	volts
1127	1227	1327	1427	270	10	100
1128	1228	1328	1428	330	10	100
1129	1229	1329	1429	330	20	100
1130	1230	1330	1430	390	10	100
1131	1231	1331	1431	470	10	100
1132	1232	1332	1432	470	20	100
1133	1233	1333	1433	560	10	100
1134	1234	1334	1434	680	10	100
1135	1235	1335	1435	680	20	100
1136	1236	1336	1436	820	10	100
1137	1237	1337	1437	1,000	10	100
1138	1238	1338	1438	1,000	20	100
1139	1239	1339	1439	1,200	10	100
1140	1240	1340	1440	1,500	10	100
1141	1241	1341	1441	1,500	20	100
1142	1242	1342	1442	1,800	10	100
1143	1243	1343	1443	2,200	10	100
1144	1244	1344	1444	2,200	20	100
1145	1245	1345	1445	2,700	10	100
1146	1246	1346	1446	3,300	10	100
1147	1247	1347	1447	3,300	20	100
1148	1248	1348	1448	3,900	10	100
1149	1249	1349	1449	4,700	10	100
1150	1250	1350	1450	4,700	20	100
1151	1251	1351	1451	5,600	10	50
1152	1252	1352	1452	6,800	10	50
1153	1253	1353	1453	6,800	20	50
1154	1254	1354	1454	8,200	10	50
1155	1255	1355	1455	10,000	10	50
1156	1256	1356	1456	10,000	20	50
Style CKR17, voltage-temperature limits of ±15% and +15, -25%						
0397	0457	0517	0577	5,600	10	100
0398	0458	0518	0578	6,800	10	100
0399	0459	0519	0579	6,800	20	100
0400	0460	0520	0580	8,200	10	100
0401	0461	0521	0581	10,000	10	100
0402	0462	0522	0582	10,000	20	100
1163	1263	1363	1463	12,000	10	50
1164	1264	1364	1464	15,000	10	50
1165	1265	1365	1465	15,000	20	50
1166	1266	1366	1466	18,000	10	50
1167	1267	1367	1467	22,000	10	50
1168	1268	1368	1468	22,000	20	50
Style CKR18, voltage-temperature limits of ±15% and +15, -25%						
1169	1269	1369	1469	12,000	10	100
0403	0463	0523	0583	15,000	10	100
0404	0464	0524	0584	15,000	20	100
1172	1272	1372	1472	18,000	10	100
0405	0465	0525	0585	22,000	10	100
0406	0466	0526	0586	22,000	20	100
0407	0467	0527	0587	27,000	10	100
0408	0468	0528	0588	33,000	10	100
0409	0469	0529	0589	33,000	20	100
0410	0470	0530	0590	39,000	10	100
0411	0471	0531	0591	47,000	10	100

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(B) TABLE I. Dash number and electrical characteristics - Continued.

Dash number				Capacitance	Capacitance tolerance	DC rated voltage
Failure rate level (%/1,000 hours)						
1.0(M)	0.1(P)	0.01(R)	0.001(S)			
Style CKR18, voltage-temperature limits of ±15% and +15, -25%						
				pF	±percent	volts
0412	0472	0532	0592	47,000	20	100
0413	0473	0533	0593	56,000	10	100
0414	0474	0534	0594	68,000	10	100
0415	0475	0535	0595	68,000	20	100
0416	0476	0536	0596	82,000	10	100
0417	0477	0537	0597	100,000	10	100
0418	0478	0538	0598	100,000	20	100
1187	1287	1387	1487	120,000	10	50
1188	1288	1388	1488	150,000	10	50
1189	1289	1389	1489	150,000	20	50
1190	1290	1390	1490	180,000	10	50

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(B)

TABLE II. Substitutability data.

Dash numbers in: MIL-C-39014/13B 1/				Substitutable for dash numbers in: MIL-C-39014/13A 1/					Substitutable for dash numbers in: MIL-C-39014/13				
Style CKR20				Style CKR17									
M	P	R	S	L	M	P	R	S	L	M	P	R	S
1101	1201	1301	1401	0301	0361	0421	0481	0541	0001	0061	0121	0181	0241
1102	1202	1302	1402	0302	0362	0422	0482	0542	0002	0062	0122	0182	0242
1103	1203	1303	1403	----	----	----	----	----	----	----	----	----	----
1104	1204	1304	1404	0303	0363	0423	0483	0543	0003	0063	0123	0183	0243
1105	1205	1305	1405	0304	0364	0424	0484	0544	0004	0064	0124	0184	0244
1106	1206	1306	1406	----	----	----	----	----	----	----	----	----	----
1107	1207	1307	1407	0305	0365	0425	0485	0545	0005	0065	0125	0185	0245
1108	1208	1308	1408	0306	0366	0426	0486	0546	0006	0066	0126	0186	0246
1109	1209	1309	1409	----	----	----	----	----	----	----	----	----	----
1110	1210	1310	1410	0307	0367	0427	0487	0547	0007	0067	0127	0187	0247
1111	1211	1311	1411	0308	0368	0428	0488	0548	0008	0068	0128	0188	0248
1112	1212	1312	1412	----	----	----	----	----	----	----	----	----	----
1113	1213	1313	1413	0309	0369	0429	0489	0549	0009	0069	0129	0189	0249
1114	1214	1314	1414	0310	0370	0430	0490	0550	0010	0070	0130	0190	0250
1115	1215	1315	1415	----	----	----	----	----	----	----	----	----	----
1116	1216	1316	1416	0311	0371	0431	0491	0551	0011	0071	0131	0191	0251
1117	1217	1317	1417	0312	0372	0432	0492	0552	0012	0072	0132	0192	0252
1118	1218	1318	1418	----	----	----	----	----	----	----	----	----	----
1119	1219	1319	1419	0313	0373	0433	0493	0553	0013	0073	0133	0193	0253
1120	1220	1320	1420	0314	0374	0434	0494	0554	0014	0074	0134	0194	0254
1121	1221	1321	1421	----	----	----	----	----	----	----	----	----	----
1122	1222	1322	1422	0315	0375	0435	0495	0555	0015	0075	0135	0195	0255
1123	1223	1323	1423	0316	0376	0436	0496	0556	0016	0076	0136	0196	0256
1124	1224	1324	1424	----	----	----	----	----	----	----	----	----	----
1125	1225	1325	1425	0317	0377	0437	0497	0557	0017	0077	0137	0197	0257
1126	1226	1326	1426	0318	0378	0438	0498	0558	0018	0078	0138	0198	0258
1127	1227	1327	1427	----	----	----	----	----	----	----	----	----	----
1128	1228	1328	1428	0319	0379	0439	0499	0559	0019	0079	0139	0199	0259
1129	1229	1329	1429	0320	0380	0440	0500	0560	0020	0080	0140	0200	0260
1130	1230	1330	1430	----	----	----	----	----	----	----	----	----	----
1131	1231	1331	1431	0321	0381	0441	0501	0561	0021	0081	0141	0201	0261
1132	1232	1332	1432	0322	0382	0442	0502	0562	0022	0082	0142	0202	0262
1133	1233	1333	1433	----	----	----	----	----	----	----	----	----	----
1134	1234	1334	1434	0323	0383	0443	0503	0563	0023	0083	0143	0203	0263
1135	1235	1335	1435	0324	0384	0444	0504	0564	0024	0084	0144	0204	0264
1136	1236	1336	1436	----	----	----	----	----	----	----	----	----	----
1137	1237	1337	1437	0325	0385	0445	0505	0565	0025	0085	0145	0205	0265
1138	1238	1338	1438	0326	0386	0446	0506	0566	0026	0086	0146	0206	0266
1139	1239	1339	1439	----	----	----	----	----	----	----	----	----	----
1140	1240	1340	1440	0327	0387	0447	0507	0567	0027	0087	0147	0207	0267

1/ Dash numbers in lower FR level columns are substitutable for dash numbers in all higher FR level columns.

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(B) TABLE II. Substitutability data - Continued.

Dash numbers in: MIL-C-39014/13B 1/				Substitutable for dash numbers in: MIL-C-39014/13A 1/					Substitutable for dash numbers in: MIL-C-39014/13				
Style CKR20				Style CKR17									
M	P	R	S	L	M	P	R	S	L	M	P	R	S
1141	1241	1341	1441	0328	0388	0448	0508	0568	0028	0088	0148	0208	0268
1142	1242	1342	1442	----	----	----	----	----	----	----	----	----	----
1143	1243	1343	1443	0329	0389	0449	0509	0569	0029	0089	0149	0209	0269
1144	1244	1344	1444	0330	0390	0450	0510	0570	0030	0090	0150	0210	0270
1145	1245	1345	1445	0331	0391	0451	0511	0571	0031	0091	0151	0211	0271
1146	1246	1346	1446	0332	0392	0452	0512	0572	0032	0092	0152	0212	0272
1147	1247	1347	1447	0333	0393	0453	0513	0573	0033	0093	0153	0213	0273
1148	1248	1348	1448	0334	0394	0454	0514	0574	0034	0094	0154	0214	0274
1149	1249	1349	1449	0335	0395	0455	0515	0575	0035	0095	0155	0215	0275
1150	1250	1350	1450	0336	0396	0456	0516	0576	0036	0096	0156	0216	0276
1151	1251	1351	1451										
1152	1252	1352	1452										
1153	1253	1353	1453										
1154	1254	1354	1454										
1155	1255	1355	1455										
1156	1256	1356	1456										
STYLE CKR17													
0397	0457	0517	0577	0337	0397	0457	0517	0577	0037	0097	0157	0217	0277
0398	0458	0518	0578	0338	0398	0458	0518	0578	0038	0098	0158	0218	0278
0399	0459	0519	0579	0339	0399	0459	0519	0579	0039	0099	0159	0219	0279
0400	0460	0520	0580	0340	0400	0460	0520	0580	0040	0100	0160	0220	0280
0401	0461	0521	0581	0341	0401	0461	0521	0581	0041	0101	0161	0221	0281
0402	0462	0522	0582	0342	0402	0462	0522	0582	0042	0102	0162	0222	0282
1163	1263	1363	1463										
1164	1264	1364	1464										
1165	1265	1365	1465										
1166	1266	1366	1466										
1167	1267	1367	1467										
1168	1268	1368	1468										
STYLE CKR18													
1169	1269	1369	1469	----	----	----	----	----	----	----	----	----	----
0403	0463	0523	0583	0343	0403	0463	0523	0583	0043	0103	0163	0223	0283
0404	0464	0524	0584	0344	0404	0464	0524	0584	0044	0104	0164	0224	0284
1172	1272	1372	1472	----	----	----	----	----	----	----	----	----	----
0405	0465	0525	0585	0345	0405	0465	0525	0585	0045	0105	0165	0225	0285
0406	0466	0526	0586	0346	0406	0466	0526	0586	0046	0106	0166	0226	0286
0407	0467	0527	0587	0347	0407	0467	0527	0587	0047	0107	0167	0227	0287
0408	0468	0528	0588	0348	0408	0468	0528	0588	0048	0108	0168	0228	0288
0409	0469	0529	0589	0349	0409	0469	0529	0589	0049	0109	0169	0229	0289
0410	0470	0530	0590	0350	0410	0470	0530	0590	0050	0110	0170	0230	0290
0411	0471	0531	0591	0351	0411	0471	0531	0591	0051	0111	0171	0231	0291
0412	0472	0532	0592	0352	0412	0472	0532	0592	0052	0112	0172	0232	0292
0413	0473	0533	0593	0353	0413	0473	0533	0593	0053	0113	0173	0233	0293
0414	0474	0534	0594	0354	0414	0474	0534	0594	0054	0114	0174	0234	0294
0415	0475	0535	0595	0355	0415	0475	0535	0595	0055	0115	0175	0235	0295
0416	0476	0536	0596	0356	0416	0476	0536	0596	0056	0116	0176	0236	0296
0417	0477	0537	0597	0357	0417	0477	0537	0597	0057	0117	0177	0237	0297
0418	0478	0538	0598	0358	0418	0478	0538	0598	0058	0118	0178	0238	0298
1187	1287	1387	1487										
1188	1288	1388	1488										
1189	1289	1389	1489										
1190	1290	1390	1490										

1/ Dash numbers in lower FR level columns are substitutable for dash numbers in all higher FR level columns.

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PACKAGING: Capacitors will be furnished in tape and reel packaging when so specified in the ordering data.

APPLICATION NOTES:

1. Substitutability for failure rates and capacitance tolerances shall be as follows:

Failure Rate	Will Replace Failure Rate	Cap Tolerance	Will Replace Tolerance
S	R, P, M, L	K	M
R	P, M, L		
P	M, L		
M	L		

Custodians:
Army - EL
Navy - EC
Air Force - 11
NASA - NA

Review activities:
Army - MI, AR
Navy - EC
Air Force - 17, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5910-1221)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-C-39014/13B CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), ESTABLISHED RELIABILITY, STYLES CKR20, CKR17, AND CKR18

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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